

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2: SYMMETRY AND RELATIVITY

TRINITY TERM 2019

Wednesday, 19 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

The following notation is used throughout the paper: capital bold letters (e.g., **U**) indicate 4-vectors; small case bold letters (e.g., **v**) indicate 3-vectors.

However the symbols **E** and **B** exceptionally indicate 3-vectors, being the electric and magnetic fields, respectively.

1. (a) For the worldline of a particle $X^\mu(\tau)$, define proper time, the 4-velocity $\mathbf{U}$ and the 4-acceleration $\mathbf{A}$, and show that $\mathbf{U} \cdot \mathbf{A} = 0$. [5]

(b) Consider a particle accelerated from rest under a constant 3-force $\mathbf{f} = (f_x, 0, 0)$. Derive expressions for $x(t)$, $\dot{x}(t)$ and $\ddot{x}(t)$. [4]

In each of the remaining parts of the question, you are asked to draw a spacetime diagram and show on it the sets of events that

- (i) can be causally affected by the origin $(0, 0, 0, 0)$
- (ii) can causally affect $(0, 0, 0, 0)$
- (iii) can neither cause nor be caused by $(0, 0, 0, 0)$.

In each case you can restrict your diagram to only one spatial coordinate x . These spacetimes are

(c) Ordinary Minkowski spacetime. Also draw the worldline of a charged particle accelerated from rest at the origin under a constant electric field $(E_x, 0, 0)$. [4]

(d) The same spacetime as in part (c), except with axes drawn in (t', x') coordinates, where (t', x') are related to (t, x) by a standard boost (you do *not* need to draw the particle worldline). [2]

(e) A spacetime where the x spatial dimension has the geometry of a circle, so that travelling a distance $2\pi R$, where R is the radius of the circle, in the x -direction returns you to your original location (you may find the identification $x \equiv x + 2\pi R$ to be a useful way of implementing this geometry).

Also draw the worldline of a charged particle accelerated from rest at the origin under a constant electric field $(E_x, 0, 0)$. Show that the particle repeatedly passes through the same point but with increased speed each time. How is this behaviour consistent with energy conservation? Justify your answer. [6]

(f) Finally, a spacetime where the temporal dimension has the geometry of a circle, with an identification $t \equiv t + 2\pi T$, where T is the radius of the circle.

For this final case, comment on your answer and whether it exhibits any inconsistencies. [4]

2. (a) The frame S' is related to the frame S by a standard Lorentz boost with velocity v in the x -direction. In frame S' a rod of length L aligned along the x' axis is moving in the positive y' direction at speed u . Determine the angle that the rod makes to the x axis in frame S .

[5]

(b) A source at the origin in frame S emits an electromagnetic signal at a constant frequency f_0 . A passing observer moves with a constant velocity $(0, v, 0)$ in the y -direction, passing through a point $(L, 0, 0)$ in S at $t = 0$. By deriving an appropriate expression for the Doppler shift, calculate and plot (as a function of t) the frequency received by the moving observer along its path.

[7]

(c) [Note: the process in this part contains a lot of physics. Full analytic expressions and derivations are not required, and qualitative/semi-quantitative answers can obtain full marks. However, you are expected to demonstrate an understanding of the key physical effects that will determine the final results.]

We now suppose that, in the configuration in part (b), the passing ‘observer’ is instead a large cloud of initially unexcited atoms, which have a single atomic absorption/emission transition at a certain frequency f and with an intrinsic lifetime τ . Starting at time $t = -T \ll 0$ when the cloud is very far away, the source at the origin emits a spherically symmetric pattern of radiation at a frequency $f_0 < f$. Some of this radiation will be absorbed by the passing atoms, placing them in an excited state. These atoms then re-emit as they decay back down to the ground state. Plot, with key features labelled and explained, how both the amplitude and frequency of the radiation received *at the source at the origin* will depend on time. Your answer should

(i) include separate plots for at least three different velocities, ranging from the non-relativistic limit $v \ll c$ to the ultra-relativistic limit $v \sim c$ and including any critical values for v that lie inbetween,

(ii) explain qualitatively the effect of the ratio f_0/f ,

(iii) take into account the finite width of the transition.

You may assume throughout that atomic absorption/emission processes do not alter the atoms’ speed v .

[13]

3. You may treat $\epsilon_0 = \mu_0 = c = 1$ for the purpose of this question.

(a) What is the difference between an *invariant quantity* and a *conserved quantity* in the context of relativity? Explain how 4-vectors and tensors transform under Lorentz transformations. [4]

(b) Show that if a 4-vector Y^μ has a component that is zero in all frames, then $Y^\mu = 0$. Give a physical consequence of this statement. Is the same statement true for tensors? Either prove it or give a counterexample. [5]

(c) State the form of the electromagnetic 4-potential A_μ and how it relates to the field tensor $F_{\mu\nu}$. Obtain an explicit expression for $F_{\mu\nu}$ in terms of $\mathbf{E}$ and $\mathbf{B}$ fields. Using this, obtain the transformations of $\mathbf{E}$ and $\mathbf{B}$ under a Lorentz transformation. [9]

(d) Using the explicit form of $F_{\mu\nu}$, or otherwise, show that $\mathbf{E}^2 - \mathbf{B}^2$ and $\mathbf{E} \cdot \mathbf{B}$ are invariants under Lorentz transformations. Assuming that $\mathbf{E}$ and $\mathbf{B}$ fields vanish at infinity, also show that

$$\int \mathbf{E} \cdot \mathbf{B} d^4x = 0$$

for any electromagnetic field configuration. [7]

4. (a) The Klein–Gordon equation for a scalar field is

$$0 = -\partial_\mu \partial^\mu \phi + m^2 \phi.$$

Obtain both plane wave and static (time-independent) spherically symmetric solutions of the Klein–Gordon equation. What restriction applies to the plane wave solutions and what physical interpretation can you give to this restriction? [6]

(b) The static solution can be understood as a potential generated for a force mediated by a massive scalar. For the two cases of $m = m_\pi = 135 \text{ MeV c}^{-2}$ and $m = m_H = 125 \text{ GeV c}^{-2}$, by how much smaller is this potential on nuclear (10^{-15} m) and atomic (10^{-10} m) scales compared to its value at a scale of 10^{-18} m ? What would be the equivalent results for a force mediated by a massless scalar? What are the physical implications of your results for understanding (i) nuclei and (ii) molecules? [5]

(c) What is a 2-spinor? Which types of particle does it describe? [2]

(d) A 2-spinor u can be mapped to a 4-vector U^μ through the equation

$$U^\mu = \frac{1}{2} u^\dagger \sigma^\mu u,$$

where $\sigma^\mu = (\mathbf{1}, \sigma_x, \sigma_y, \sigma_z)$. Show that the vector U^μ is null. If $u \rightarrow e^{\rho\sigma_z/2} u$, show that the vector U^μ undergoes a Lorentz transformation and determine that transformation. [Hint: you may wish to use the hyperbolic form of the Lorentz transformations and the hyperbolic double angle formulae.] Based on this, what transformation would you expect $u \rightarrow e^{i\theta\sigma_z} u$ to correspond to? [10]

(e) Would you expect a 2-spinor to satisfy the massless Klein–Gordon equation? Explain your logic in as much detail as you can. [2]

[The Pauli matrices are: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.]